SIFT

1. People say 'SIFT' but there are two parts to SIFT [1].
    1. an interest point detector.
    2. a region descriptor.
        1. They are independent, many people use the region descriptor without looking for the points.
2. SIFT summary[2]:
    1. Extract a 16x16 patch around detected keypoint.
    2. Compute the gradients and apply a Gaussian weighting function.
    3. Divide the window into a 4x4 grid of cells.
    4. Compute gradient direction histograms over 8 directions in each cell.
    5. Concatenate the histograms over 8 directions in each cell.
    6. Concatenate the histograms to obtain 128 dimensional feature vector.
    7. Normalize to unit length.

	Rotation Invariant	Scale Invariant	Affine Invariant	Repeatability	Localization Accuracy	Robustness	Efficiency
Haris	x			+++	+++	++	++
Shi-Tomasi	x			+++	+++	++	++
FAST	x	x		++	++	++	++++
SIFT	x	x	x	+++	++	+++	+
SURF	x	x	x	+++	++	++	++

Image Courtesy of Davide Scaramuzza et. al 2012.

Local features: main components [3]

1. Detection: Identify the interest points.
2. Description: Extract vector feature descriptor surrounding each interest point.
3. Matching: Determine correspondence between descriptors in two views.

References:

1. CSE576 Udub Linda Shapiro
2. Trym Vegard Haavardsholm, UNIK4690
3. CSIE 5429 3D-DLCV, National Taiwan University, Spring 2021